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Deployment-Agnostic Indoor Positioning Using BLE-Based Localization in Smart Spaces

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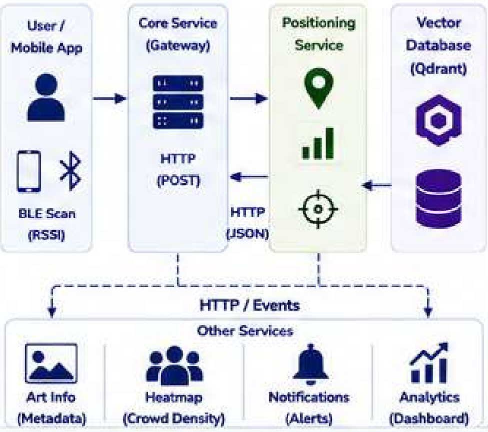
1 CHALLENGES & TRADE-OFFS

- No GPS indoors – Signals are blocked by walls and ceilings.
- Noisy RSSI signals – Affected by noise, reflection, and obstruction.
- Distance-based methods unreliable – Multipath and environment changes reduce accuracy.
- Accuracy depends on beacon placement.
- Signal model approximates real environments.

2 RESEARCH QUESTIONS

1. How to make positioning robust to noisy BLE signals?
2. How to enable deployment-agnostic positioning without training?
3. How to continuously refine fingerprints using implicit signals (e.g., NFC/QR)? (Future work)
4. How to represent uncertainty in location estimates? (Future work)

3 SYSTEM ARCHITECTURE



METHOD: BLE VECTOR SIMILARITY PIPELINE

A. CREATING THE REFERENCE (FINGERPRINT) DATABASE (OFFLINE / SETUP)

1. Define layout (grid, floors)
2. Place beacons & collect Tx Power (1m reference)
3. Configure signal model parameters (n, P0, etc.)
4. For each grid cell (x, y), estimate RSSI from all beacons
5. Form reference vector (fixed length) — B_1 -65 dBm, B_2 -72 dBm, ... , B_n -88 dBm
6. Store vector with location label (x, y) in database

REFERENCE VECTOR EXAMPLE (normalized to [0,1])

Reference (fingerprint) database: one vector per grid cell representing all possible locations.

B_1	B_2	...	B_n	Location (x,y)
0.12	-0.35	...	-0.22	(1,1)
0.10	-0.28	...	-0.21	(1,2)
-0.15	-0.32	...	-0.30	(2,1)
...	...	...	...	...

Vectors are normalized to [min-max 0,1] for fair comparison.

B. ESTIMATE THE USER'S CURRENT LOCATION (ONLINE / RUNTIME)

Goal: Match the live user vector against the reference database to estimate the most likely location.

1. Live BLE Scan (from user)
2. Construct User Vector (from RSSI): -65 dBm B_1; -68 dBm B_2; -81 dBm B_3; -72 dBm B_4; ... B_n
3. Normalize Vector [min-max 0,1]: B_1 B_2 ... B_n — 0.12 -0.35 ... -0.21
4. Compare with Reference Vectors (cosine similarity): (1,1) 0.87; (1,2) 0.72; (2,1) 0.43; ...
5. Estimated User Location: Grid (1,1) Score: 0.87

MATCHING PROCESS (DETAILS)

1. Build user vector from live RSSI observations.
2. Normalize user and reference vectors to [0,1] using min-max scaling.
3. Compute cosine similarity between user vector and each reference vector.
4. Select grid cell with highest similarity as the user's location.

- Unsupervised fingerprinting – no data collection or training required.
- Handles noisy and unstable RSSI signals.
- Low-latency response suitable for real-time applications.
- Deployment-agnostic via configuration (layout, beacons, parameters).

DEPLOYMENT-AGNOSTIC DESIGN

Same system, different environment

New SmartSpace — Deploy to any new space or building → Configure (YAML Only) — Update layout, beacons and signal parameters → Ready for Deployment — System is fully operational and adapted

What can change (configuration only)
- Beacon positions
- Layout (grid)
- Signal parameters (Tx Power, Propagation Model)

What stays constant
- Vector construction
- Normalization
- Matching algorithm
- System logic & flow

IMPLEMENTATION: REAL-TIME HEATMAP (OUR SYSTEM)

KEY APPLICATIONS

INDOOR NAVIGATION — Guide visitors to destinations and points of interest.

CONTEXT-AWARE INFO — Deliver location-based content and services.

LOCATION-BASED ADVERTISING — Deliver relevant offers and promotions.

RELATED WORK

Rhaled, A. (2024). Smart Location-Based Services (Smart-LBS): Platform for Smart Space-Independent LBS.

BLE FINGERPRINTING | VECTOR SIMILARITY | DEPLOYMENT-AGNOSTIC | REAL-TIME | SCALABLE